COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	16-07-2026	**DATE SUBMITTED:**	16-07-2026
NAME:	Dueñas, Ranzel Aldous Gabriel L.	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 1.1 **Creating Virtual Machines**

1. PROCEDURE OUTPUT

Task 1: Changing Hostname

1. Change the hostname using the command sudo nano /etc/hostname

1.1 Use server1 for Server 1



1.2 Use server2 for Server 2

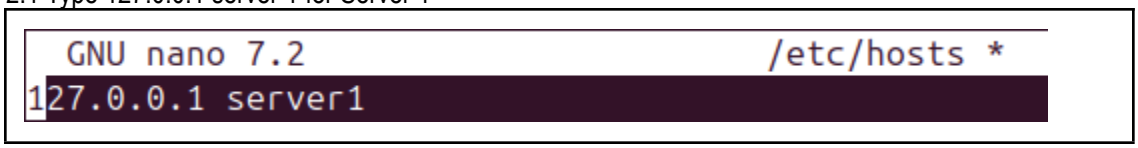


1.3 Use workstation for the Local Machine



2. Edit the hosts using the command sudo nano /etc/hosts. Edit the second line.

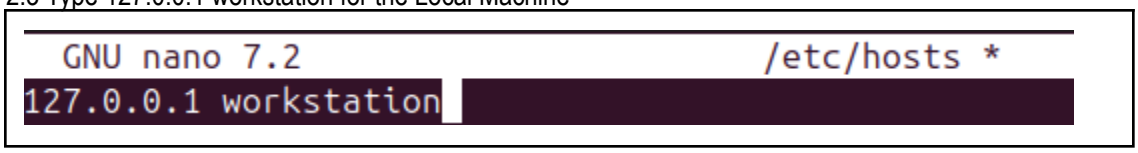
2.1 Type 127.0.0.1 server 1 for Server 1



2.2 Type 127.0.0.1 server 2 for Server 2



2.3 Type 127.0.0.1 workstation for the Local Machine



Task 2: Configure SSH

Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

1. Upgrade the packages by issuing the commands sudo apt update and sudo apt upgrade, respectively.

```
duenas@workstation:~$ sudo apt update && sudo apt upgrade
[sudo] password for duenas:
Hit:1 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
duenas@workstation:~$
```

```
duenas@server1:~$ sudo apt update && sudo apt upgrade
[sudo] password for duenas:
Sorry, try again.
[sudo] password for duenas:
Hit:1 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Fetched 126 kB in 1s (93.7 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
duenas@server1:~$
```

```
duenas@server2:~$ sudo apt update && sudo apt upgrade
Hit:1 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
duenas@server2:~$
```

2. Install the SSH server using the command sudo apt install openssh-server.

```
duenas@workstation:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
duenas@server1:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
```

```
duenas@server2:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

3. Verify if the SSH service has started by issuing the following commands:

3.1 sudo service ssh start

```
duenas@workstation:~$ sudo service ssh start
```

```
duenas@server1:~$ sudo service ssh start
```

```
duenas@server2:~$ sudo service ssh start
```

3.2 sudo systemctl status ssh

```
duenas@workstation:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: ena>
     Active: active (running) since Thu 2026-07-16 01:20:35 UTC; 1min 52s ago
TriggeredBy: ● ssh.socket
```

```
duenas@server1:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: ena>
     Active: active (running) since Thu 2026-07-16 01:21:02 UTC; 2min 22s ago
TriggeredBy: ● ssh.socket
```

```
duenas@server2:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: ena>
     Active: active (running) since Thu 2026-07-16 01:20:50 UTC; 2min 51s ago
TriggeredBy: ● ssh.socket
```

4. Configure the firewall to all port 22 by issuing the following commands:

4.1 sudo ufw allow ssh

```
duenas@workstation:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
```

```
duenas@server1:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
```

```
duenas@server2:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
```

4.2 sudo ufw enable

```
duenas@workstation:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

```
duenas@server1:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

```
duenas@server2:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

4.3 sudo ufw status

```
duenas@workstation:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

```
duenas@server1:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

```
duenas@server2:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

Task 3: Verify network settings on Server 1, Server 2, and Local Machine.

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command ifconfig and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.
   - 1.1 Server 1 IP address: 192.168.56.107
   - 1.2 Server 2 IP address: 192.168.56.108
   - 1.3 Server 3 IP address: 192.168.56.109
2. Make sure that they can ping each other.
   - 2.1 Connectivity test for Local Machine 1 to Server 1: ■ Successful ☐ Not Successful
   - 2.2 Connectivity test for Local Machine 1 to Server 2: ■ Successful ☐ Not Successful
   - 2.3 Connectivity test for Server 1 to Server 2: ■ Successful ☐ Not Successful

```
duenas@workstation:~$ ping 192.168.56.108
PING 192.168.56.108 (192.168.56.108) 56(84) bytes of data.
64 bytes from 192.168.56.108: icmp_seq=1 ttl=64 time=1.17 ms
64 bytes from 192.168.56.108: icmp_seq=2 ttl=64 time=0.650 ms
64 bytes from 192.168.56.108: icmp_seq=3 ttl=64 time=0.460 ms
64 bytes from 192.168.56.108: icmp_seq=4 ttl=64 time=0.454 ms
64 bytes from 192.168.56.108: icmp_seq=5 ttl=64 time=0.409 ms
^Z
[1]+  Stopped                 ping 192.168.56.108
duenas@workstation:~$ ping 192.168.56.109
PING 192.168.56.109 (192.168.56.109) 56(84) bytes of data.
64 bytes from 192.168.56.109: icmp_seq=1 ttl=64 time=0.831 ms
64 bytes from 192.168.56.109: icmp_seq=2 ttl=64 time=0.748 ms
64 bytes from 192.168.56.109: icmp_seq=3 ttl=64 time=0.480 ms
64 bytes from 192.168.56.109: icmp_seq=4 ttl=64 time=0.541 ms
^Z
[2]+  Stopped                 ping 192.168.56.109
duenas@workstation:~$
```

```
duenas@server1:~$ ping 192.168.56.109
PING 192.168.56.109 (192.168.56.109) 56(84) bytes of data.
64 bytes from 192.168.56.109: icmp_seq=1 ttl=64 time=1.28 ms
64 bytes from 192.168.56.109: icmp_seq=2 ttl=64 time=0.475 ms
64 bytes from 192.168.56.109: icmp_seq=3 ttl=64 time=0.497 ms
```

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the Local Machine, issue the following commands:

1.1 ssh username@ip_address_server1 for example, ssh jvtaylar@192.168.56.120

```
duenas@workstation:~$ ssh duenas@192.168.56.108
The authenticity of host '192.168.56.108 (192.168.56.108)' can't be established.
ED25519 key fingerprint is SHA256:mTRnDWLxPAlo/UyJkKj6Rd26mWUOSw3XchH9h5iGg3E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
```

1.2 Enter the password for server 1 when prompted

```
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.56.108' (ED25519) to the list of known hosts
.
duenas@192.168.56.108's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)
```

1.3 Verify that you are in server 1. The user should be in this format user@server1.
For example, jvtaylar@server1

```
duenas@server1:~$ whoami
duenas
```

2. Logout of Server 1 by issuing the command control + D.
3. Do the same for Server 2.

```
duenas@workstation:~$ ssh duenas@192.168.56.109
The authenticity of host '192.168.56.109 (192.168.56.109)' can't be established.
ED25519 key fingerprint is SHA256:mTRnDWLxPAlo/UyJkKj6Rd26mWUOSw3XchH9h5iGg3E.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.109' (ED25519) to the list of known hosts
```

```
duenas@192.168.56.109's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)
```

```
duenas@server2:~$ whoami
duenas
```

4. Edit the hosts of the Local Machine by issuing the command sudo nano/etc/hosts. Below all texts type the following:

4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)
4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)

```
duenas@workstation: ~
  GNU nano 7.2                    /etc/hosts *
192.168.56.108 server1
192.168.56.109 server2
```

4.3 Save the file and exit.

5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do ssh jvtaylar@server1. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2

```
duenas@workstation:~$ ssh duenas@server1
The authenticity of host 'server1 (192.168.56.108)' can't be established.
ED25519 key fingerprint is SHA256:mTRnDWLxPAlo/UyJkKj6Rd26mWUOSw3XchH9h5iGg3E.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server1' (ED25519) to the list of known hosts.
duenas@server1's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 16 02:03:52 2026 from 192.168.56.107
duenas@server1:~$
```

```
duenas@workstation:~$ ssh duenas@server2
The authenticity of host 'server2 (192.168.56.109)' can't be established.
ED25519 key fingerprint is SHA256:mTRnDWLxPAlo/UyJkKj6Rd26mWUOSw3XchH9h5iGg3E.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
    ~/.ssh/known_hosts:5: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server2' (ED25519) to the list of known hosts.
duenas@server2's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 16 02:00:23 2026 from 192.168.56.107
duenas@server2:~$
```

COMPUTER ENGINEERING

QUEZON CITY

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?
   - By putting the server IP / name to the workstation's /etc/hosts. By that the system recognizes the hostname as their IPs.
2. How secured is SSH?
   - SSH encrypts all data sent between the client and the server. Also, instead of passwords, it supports cryptographic keys, making it almost impossible to brute-force. SSH also utilizes HMAC to ensure that data cannot be altered.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.